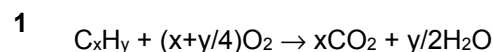


Paper 1 Worked Solutions

1	A	2	C	3	D	4	D	5	C
6	B	7	D	8	B	9	A	10	D
11	B	12	A	13	D	14	D	15	B
16	C	17	C	18	B	19	A	20	B
21	C	22	C	23	D	24	A	25	A
26	C	27	D	28	A	29	B	30	A



$$V_{O_2(excess)} + V_{CO_2} = 55$$

$$\text{Since } V_{CO_2} = 40 \text{ cm}^3, V_{O_2(excess)} = 55 - 40 = 15 \text{ cm}^3$$

$$V_{O_2(reacted)} = 80 - 15 = 65 \text{ cm}^3$$

$$\text{Mole ratio of } C_xH_y : CO_2 = 1:x = 10:40 \Rightarrow x = 4$$

$$\text{Mole ratio of } C_xH_y : O_2 = 1:(x+y/4) = 10:65 \Rightarrow y = 10$$

Formula of the hydrocarbon is C_4H_{10} .

Ans: A

- 2 Given 90% in each of the three steps,
 1 mol of $Na_2SO_4(s)$ yields 0.9 mol of $Na_2S(s)$.
 In turn, 0.9 mol of $Na_2S(s)$ yields 0.81 mol of $Na_2CO_3(s)$.
 In turn, 0.81 mol of $Na_2CO_3(s)$ yields 1.458 mol of $NaHCO_3(s)$.

$$\Rightarrow \text{Mole ratio of } Na_2SO_4(s) : NaHCO_3(s) = 1:1.458$$

$$\text{Mass of } NaHCO_3(s) = \left(\frac{100000}{142} \times 1.458 \right) \times 84.0 = 86248 \text{ g} = 86.2 \text{ kg}$$

Ans: C

- 3 Option 1 is incorrect. $^{58}Ni^{2+}$ ion has 58 nucleons while $^{60}Ni^{2+}$ ion has 60 nucleons.
 Option 2 is correct. Both Ni isotopes have 28 protons.
 Option 3 is correct. $^{58}Ni^{2+}$ ion has 30 neutrons and 26 electrons while $^{60}Ni^{2+}$ ion has 32 neutrons and 26 electrons.
 Option 4 is correct. Angle of deflection is directly proportional to $\frac{\text{charge}}{\text{mass}}$ ratio. Since $^{58}Ni^{2+}$ ion has a smaller mass than that of $^{60}Ni^{2+}$ ion, it has a

bigger angle of deflection.

Ans: D

4 $V_1P_1 + V_2P_2 = V_T P_T$
 $(4)(200) + (3)(400) = P_T(16)$
 $P_T = 125 \text{ kPa}$

Ans: D

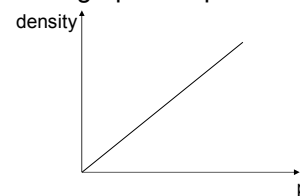
- 5 Option C is incorrect.

$$pV = nRT$$

$$p = \rho \cdot \frac{RT}{Mr}$$

$$\rho = p \cdot \frac{Mr}{RT}$$

A plot of density against pressure, at constant temperature, would give a linear graph of a positive gradient.



Ans: C

- 6 Option A is incorrect. From the given diagram, the presence of hexagons and pentagons imply different bond angles within buckminsterfullerene.
 Option B is correct. From the given diagram, all the carbon in buckminsterfullerene is sp^2 hybridised hence the 4th electron of each carbon is delocalised within the sphere.
 Option C is incorrect. The sublimation point of buckminsterfullerene should be lower than that of graphite, as it has a simple molecular structure with instantaneous dipole-induced dipole interactions between molecules whereas graphite has a giant molecular structure with strong covalent bonds between atoms.

Option D is incorrect. The absence of hydrogen in buckminsterfullerene implies the absence of water during combustion.

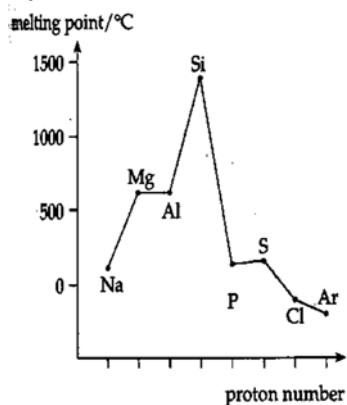
Ans: B

- 7 Option A is incorrect. Compounds can dissolve endothermically hence it is possible that barium fluoride can dissolve in water.
 Option B is incorrect. Ba^{2+} is bigger in size than Mg^{2+} hence Ba^{2+} has a lower charge density and therefore less polarising. The hydration energy of Ba^{2+} will be smaller in magnitude than that of Mg^{2+} .
 Option C is incorrect. LE depends on the charge and size of the ions. Since Ba^{2+} is bigger in size than Mg^{2+} , LE of BaF_2 will be smaller in magnitude than MgF_2 .

Option D is correct. Both ΔH_{LE} and $\Delta H_{\text{hydration}}$ are negative and $\Delta H_{\text{solution}} = \Sigma \Delta H_{\text{hydration}} - \Delta H_{\text{LE}}$. For $\Delta H_{\text{solution}}(\text{BaF}_2)$ to be negative, ΔH_{LE} must be bigger in magnitude than $\Sigma \Delta H_{\text{hydration}}$.

Ans: D

- 8 Option A is correct: Atomic radius decreases across the period. Nuclear charge increases as number of protons increases across the period. Electrons are added to the same valence shell hence shielding effect remains almost constant. Effective nuclear charge increases and thus, the valence electrons are more strongly attracted to the nucleus.
 Option B is incorrect: Melting point trend should be as shown instead, where the melting point of Si is the highest.



Option C is correct: The ability to conduct electricity depends on the presence of mobile charge carriers. Electrical conductivity increases

from Na to Al due to the increase in number of valence electrons available in "sea" of delocalised electrons. Si is a semiconductor, while P, S and Cl are non-conductors.

Option D is correct: Electronegativity increases across the period. Nuclear charge increases. Shielding effect remains relatively constant as the number of inner shell electrons remains the same. There is a larger effective nuclear charge, and thus a stronger net attraction for a bond pair of electrons in a covalent bond.

Ans: B

- 9 A halogen that is a stronger oxidising agent can oxidise a halide below it in the group. From the results of the table, it can be deduced that the oxidising power of the halogens: $\text{Y}_2 > \text{Z}_2 > \text{X}_2$
 Thus the reducing power of the halide ions is: $\text{X}^- > \text{Z}^- > \text{Y}^-$

Ans: A

- 10 From the successive IE, there is a big jump between the 5th IE and 6th IE. Hence, the 6th electron must have been removed from an inner principal quantum shell. Hence there are 5 electrons in the outermost shell and element Y is from Group 15.
 It can be inferred therefore that X: Group 13; W: Group 14; Z: Group 16

Option A is correct: The oxide of X (i.e. Al_2O_3) is insoluble in water. The small and highly charged Al^{3+} ion results in highly exothermic lattice energy. Hence, the energy that is released during hydration cannot compensate for the larger amount of energy that is required to overcome the strong ionic bonds in the giant lattice.

Option B is correct. The first IE of Z (i.e. sulfur) is lower than that of Y (i.e. phosphorous). This is one of the two exceptions in the 1st IE trends - for Z, electron is removed from a paired electron occupying the same 3p orbital. Thus, the electron experiences inter-electronic repulsion and lesser energy is required to remove it.

Option C is correct. Oxide of X (i.e. Al_2O_3) is amphoteric, and will react with both acids and bases. $\text{Al}_2\text{O}_3(\text{s}) + 6\text{HCl}(\text{aq}) \rightarrow 2\text{AlCl}_3(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$

Option D is incorrect. The chloride of W (i.e. CCl_4) does not undergo hydrolysis in water as it does not have energetically accessible vacant 3d orbitals (unlike the chlorides of Y and Z) which can accommodate lone pair of electrons from oxygen atoms of water during hydrolysis.

Ans: D

- 11 Option 1 is incorrect.

$-\Delta H_1$ = The enthalpy change of formation ΔH_f , of Be_3N_2

Since $3 \text{ Be(s)} + 3 \text{ N}_2(\text{g}) \rightarrow \text{Be}_3\text{N}_2$

$$-\Delta H_1 = \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5 - \Delta H_6$$

$$= +324 + 944 (\text{BE of N}\equiv\text{N}) + 3(900+1760) + 1404 - 11300$$

$$= -648 \text{ kJ mol}^{-1}$$

Option 2 is correct. Enthalpy change of atomisation is always endothermic. In this question, energy is absorbed to break $\text{N}\equiv\text{N}$ to produce 2 N atoms.

Option 3 is correct. ΔH_6 represents $-[\text{Lattice Energy of Be}_3\text{N}_2]$. Lattice energy is a measure of the ionic bond strength.

Ans: B

- 12 Option 1 is incorrect. $\text{S(s)} + \text{O}_2(\text{g}) \rightarrow \text{SO}_2(\text{g})$

There is no change in the number of moles of gaseous reactants and products.

Option 2 is incorrect. When a solute dissolves in solvent, there is an increase in the disorderliness of the system.

Option 3 is correct. Formation of a solid ionic compound from its constituent gaseous ions results in a decrease in disorderliness of the system as the ions are now held in fixed positions.

Ans: A

- 13 Percentage efficiency

$$= \frac{\text{heat transferred to calorimeter}}{\text{heat released due to the combustion of hydrazine}} \times 100\%$$

$$= \frac{\Delta T \times C}{\text{amount of hydrazine used} \times \text{enthalpy change of combustion}} \times 100\%$$

$$= \frac{W \times Z}{\frac{225 \times 1000 \times 1.02}{\text{Molar mass (N}_2\text{H}_4)}} \times 622.2 \times 100\%$$

Ans: D

14. Option 1 is incorrect. An increase in pressure will not result in a shift in position of equilibrium since the number of moles of gaseous reactants = number of moles of gaseous products.

Option 2 is incorrect. Addition of water results in a decrease in concentration of **X**. Hence, position of equilibrium will shift to the left in order to increase the concentration of **X**. Hence, concentration of **Z** should decrease.

Option 3 is incorrect. An increase in temperature will result in the position of equilibrium shifting left in order to absorb heat energy since the backward reaction is endothermic. Hence, concentration of **Z** should decrease.

Answer: D

$$15 \text{ Since } P_Q = \frac{n_Q}{n_T} \times P_T \quad \text{and } P_R = \frac{n_R}{n_T} \times P_T$$

$$K_p = \frac{(P_R)^2}{(P_Q)^2} = \frac{(n_R)^2}{(n_Q)^2} = 64$$

$$\frac{n_R}{n_Q} = 8$$

$$n_Q : n_R = 1 : 8$$

Ans: B

- 16 Option A is incorrect.

From the graph (increasing pH as volume of Y added increases), it can be inferred that a base (**Y**) was added to an acid (**X**). **Y** cannot be HCl . Option B is incorrect. Even though **Y** contains 2 strong bases, the total $[\text{OH}^-]$ is still 0.1 mol dm^{-3} , hence 10 cm^3 of **Y** is needed to neutralise **X**.

Option C is correct. Na_2CO_3 is dibasic. A simple balanced chemical equation (carbonates react with acids to give salt, water and carbon dioxide) will prove that $\text{Na}_2\text{CO}_3 : \text{HCl}$ is 2 : 1 and thus only 5 cm^3 of Na_2CO_3 is needed to react with HCl .

Option D is incorrect. NaHCO_3 is monobasic and 10 cm^3 of NaHCO_3 is required to neutralise HCl .

Ans: C

- 17 Option 1 is incorrect. When temperature increases, the curve is displaced right and the peak is lowered as the total number of molecules remains constant. Proportion of molecules does not increase at all energies.

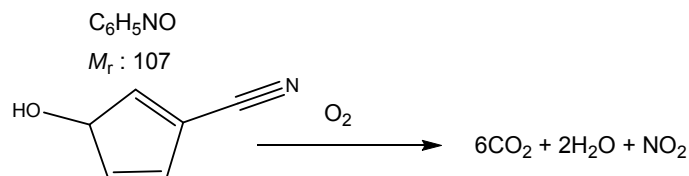
Option 2 is correct. When temperature increases, the curve is displaced right and peak of graph is lowered as the total number of molecules

remains constant.

Option 3 is correct. When temperature is increased, the value of rate constant k increases.

Ans: C

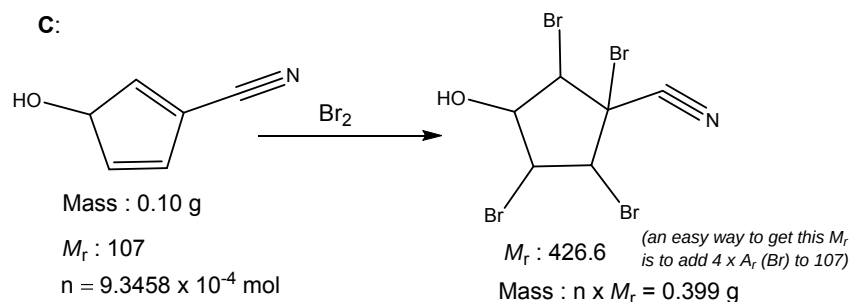
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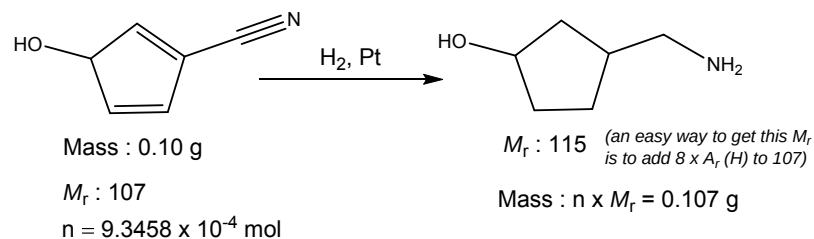
A: Mass of CO_2 formed : $6 \times 9.3458 \times 10^{-4} \times 44 = 0.247 \text{ g}$

B: Volume of NO_2 formed : $9.3458 \times 10^{-4} \times 24000 = 22.4 \text{ cm}^3$

C:



D:



Ans: B

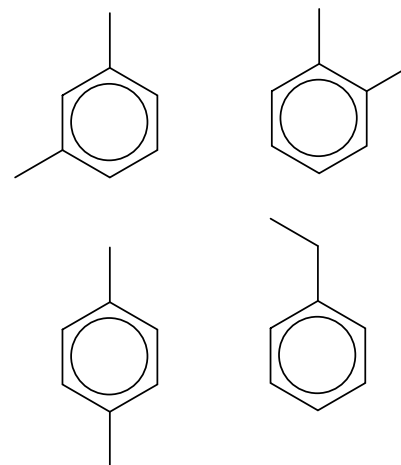
- 19 Option 1: CN^- can approach the planar carbonyl carbon from top and bottom with equal probability to give enantiomers (with chiral carbon).
 Option 2: Even though a tertiary carbocation is formed as intermediate

via the $\text{S}_{\text{N}}1$ mechanism, the final product does not have a chiral carbon, hence no enantiomers are formed.

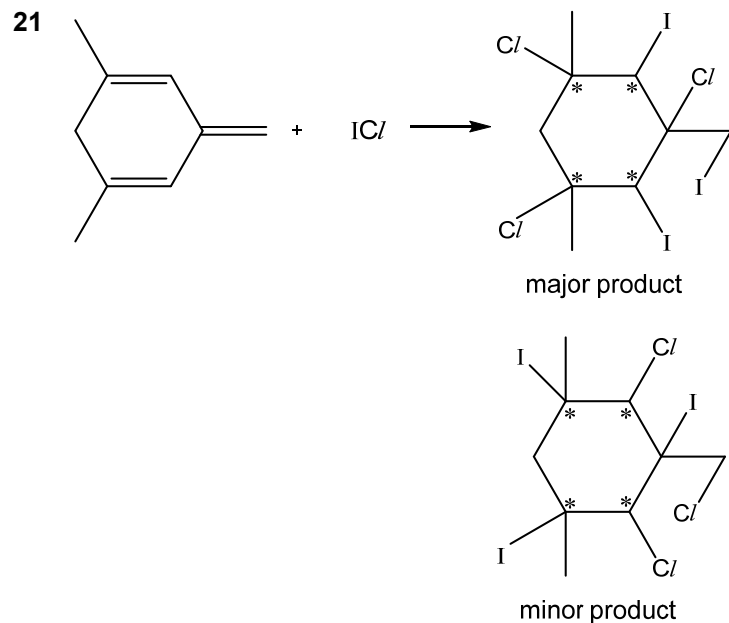
Option 3: $\text{S}_{\text{N}}2$ mechanism has occurred as the reactant is a primary halogenoalkane. The product formed does not have a chiral centre, thus it is optically inactive.

Ans: A

20



Ans: B

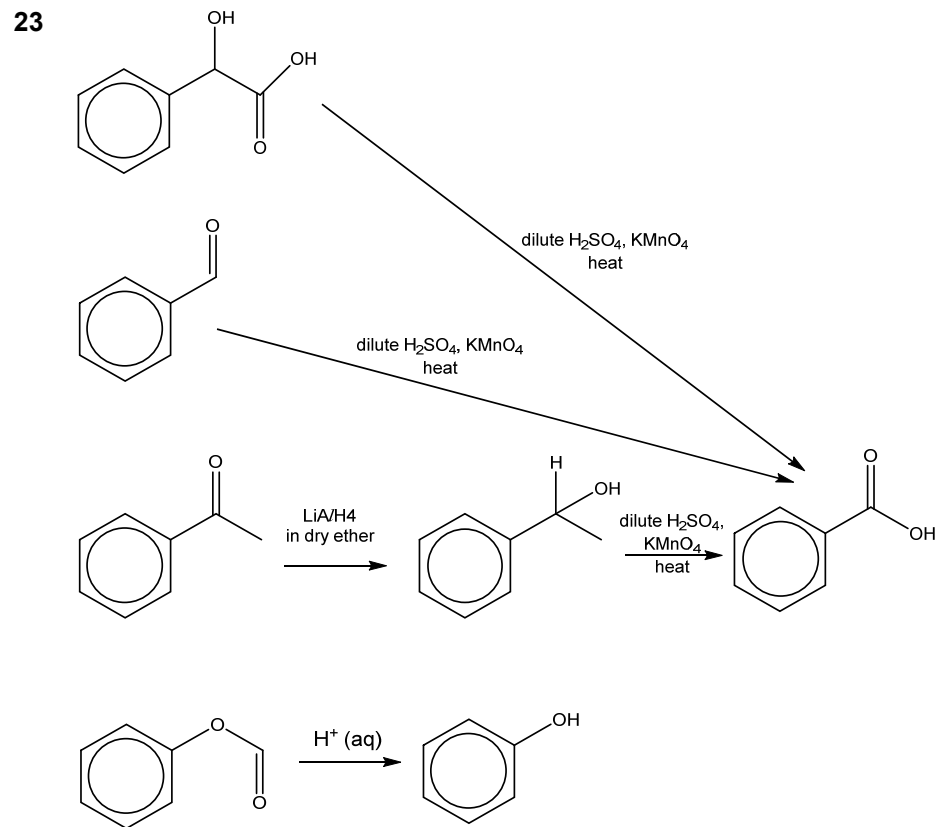


Applying Markovnikov's rule, there are 4 chiral carbons in the major product.

Ans: C

- 22 Benzene prefers to undergo substitution reactions rather than addition reactions due to delocalisation of π -electrons, which leads to resonance stability. Addition reaction destroys this stability. Options A, B and D are true but does not explain why benzene undergoes substitution instead of addition.

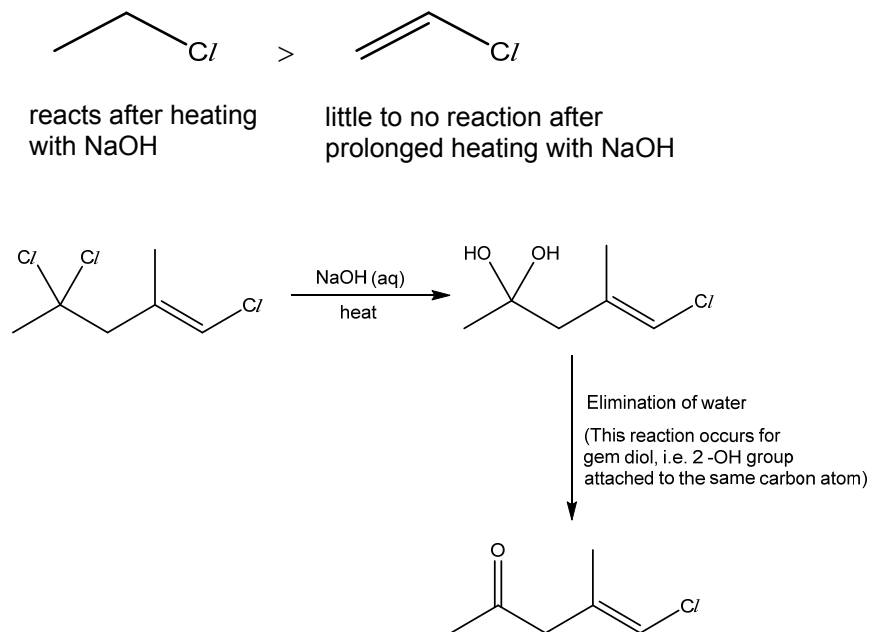
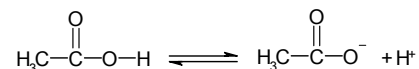
Ans: C



Ans: D

- 24 The following reactions occur:
- Reduction of nitrogen oxides to nitrogen and oxygen with rhodium catalyst: $2\text{NO}_x \rightarrow x\text{O}_2 + \text{N}_2$
 - Oxidation of carbon monoxide to carbon dioxide with platinum and palladium catalyst: $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
 - Oxidation of unburnt hydrocarbons to carbon dioxide and water with platinum and palladium catalyst.

Ans: A

25 Ease of nucleophilic substitution with water:**Ans: A****26** To discuss the strength of the acids, **compare** the **stability** of the **conjugate base** of the acids.
e.g.

When the conjugate base is more stable, the acid has a greater tendency to dissociate to form the conjugate base and H^+ ions

- The forward reaction occurs to a greater extent
- The equilibrium position lies more to the right.

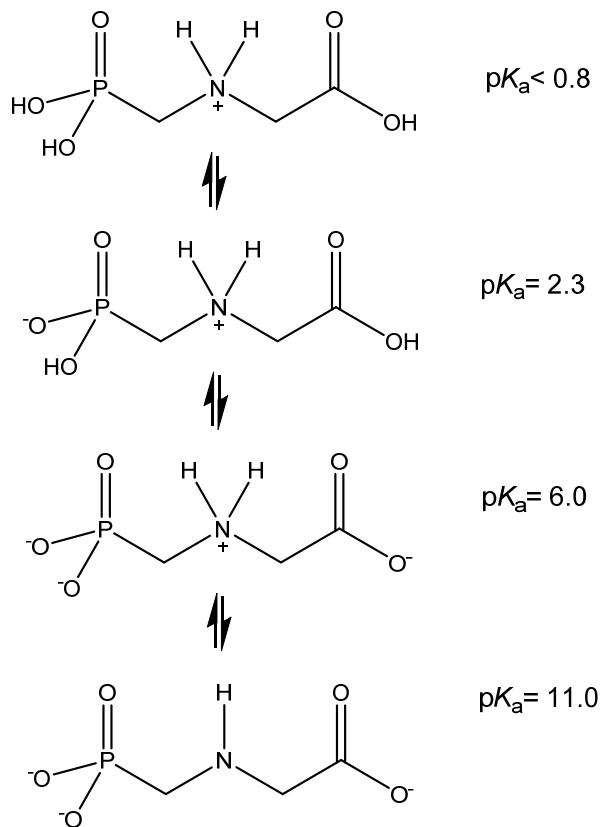
propanol	phenol	propanoic acid
$\text{ROH} + \text{H}_2\text{O} \rightleftharpoons \text{RO}^- + \text{H}_3\text{O}^+$	$\text{C}_6\text{H}_5\text{OH} + \text{H}_2\text{O} \rightleftharpoons \text{C}_6\text{H}_5\text{O}^- + \text{H}_3\text{O}^+$	$\text{RCOOH} + \text{H}_2\text{O} \rightleftharpoons \text{RCOO}^- + \text{H}_3\text{O}^+$
Alkyl groups are electron donating. Hence negative charge on the oxygen atom in RO^- is intensified compared to OH^- . RO^- is destabilised and hence alcohols are weaker acids than water.	Lone pair of electrons on the oxygen atom delocalised into the benzene ring. The negative charge on the oxygen atom in $\text{C}_6\text{H}_5\text{O}^-$ is delocalised into the ring. This dispersion of charge stabilises $\text{C}_6\text{H}_5\text{O}^-$, hence phenol is more acidic than water.	The negative charge on the oxygen atom in RCOO^- is delocalised between the 2 electronegative oxygen atoms and it is resonance stabilised. Hence carboxylic acids are more acidic than phenol.

Ans: C**27**

Information	Deduction
X , is found to decolourise aqueous bromine.	electrophilic addition $\Rightarrow$ Presence of alkene
X reacts with hot acidified potassium manganate(VII) to form compound Y .	Oxidative cleavage of alkene
Y reacts with Na_2CO_3 .	Neutralisation $\Rightarrow$ presence of carboxylic acid
Y reacts with 2,4-dinitrophenylhydrazine.	Condensation $\Rightarrow$ Presence of ketone (oxidative cleavage of alkene gives either carbon dioxide gas, carboxylic acid or ketone only)
Compound X	
Compound Y	

Ans: D

- 28 Both phosphonic acid and carboxylic acid moieties can be ionised and the amine group can be protonated and the substance exists as a series of zwitterions.



Ans: A

- 29 Option A is correct: addition of $\text{H}_2\text{O}_2(\text{aq})$ to acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ produces a green solution of Cr^{3+} .
 From Data Booklet: $\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{O}_2$ +0.68V
 $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ +1.33V
 $E_{\text{cell}} = 1.33 - (+0.68) = +0.65\text{V} > 0$, thus reaction is spontaneous.
 Option B is incorrect:

From Data Booklet: $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$ +0.54V

$\text{S}_2\text{O}_8^{2-} + 2\text{e}^- \rightleftharpoons \text{SO}_4^{2-}$ +2.01V

A suitable catalyst will have an $E^\ominus$ value in between +0.54V and +2.01V.

Since $\text{V}^{3+} + \text{e}^- \rightleftharpoons \text{V}^{2+}$ -0.26V

It is not a suitable catalyst.

Option C is correct: Ligand exchange reaction occurred. Water molecules in $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ are replaced by Cl^- ions to form $[\text{CuCl}_4]^{2-}$ accounting for the yellow solution.

$[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^- \rightleftharpoons [\text{CuCl}_4]^{2-} + 6\text{H}_2\text{O}$

Option D is correct: Hydrolysis reaction occurs for ions with acidic properties such as highly polarising $\text{Cr}^{3+}(\text{aq})$ due to its high charge density.

$[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + \text{H}_2\text{O} \rightleftharpoons [\text{Cr}(\text{H}_2\text{O})_5\text{OH}]^{2+} + \text{H}_3\text{O}^+$

When Na_2CO_3 is added: $\text{CO}_3^{2-} + 2\text{H}_3\text{O}^+ \rightarrow \text{CO}_2 + 3\text{H}_2\text{O}$

The CO_2 evolved gives a white ppt with limewater.

Ans: B

- 30 Option 1 is correct. Lattice energy is the heat released when 1 mole of MgCl_2 is formed from its constituent gaseous ions i.e. $-2526 \text{ kJ mol}^{-1}$.

Option 2 is correct. As seen from the diagram, $\Delta H_{\text{hydration}}$ of Cl^- is -384 kJ mol^{-1} .

Option 3 is correct.

$\Delta H_{\text{solution}} = 2526 + [-1890 + 2(-384)] = -132 \text{ kJ mol}^{-1}$.

Option 4 is correct. The higher energy level, the less thermally stable a species is.

Ans: A